

School mapping in ESRI imagery

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Recap from last meeting

Non-school sampling

- Previously:
 - Non-schools sampled throughout Vietnam
 - Non-schools have no connection to the splits' schools
- **Distance-based non-school sampling**
 - Sample non-schools based on distance from schools
 - **Goal:** train model only on non-school images geographically close to corresponding school images

Inverse non-school sampling

- Schools are split into two clusters
- For each cluster, calculate the corresponding **centroid**
- For each non-school (OSM Vietnam data), calculate the **Euclidean distance from both centroids**

$$d((x, y), (a, b)) = \sqrt{(x - a)^2 + (y - b)^2}$$

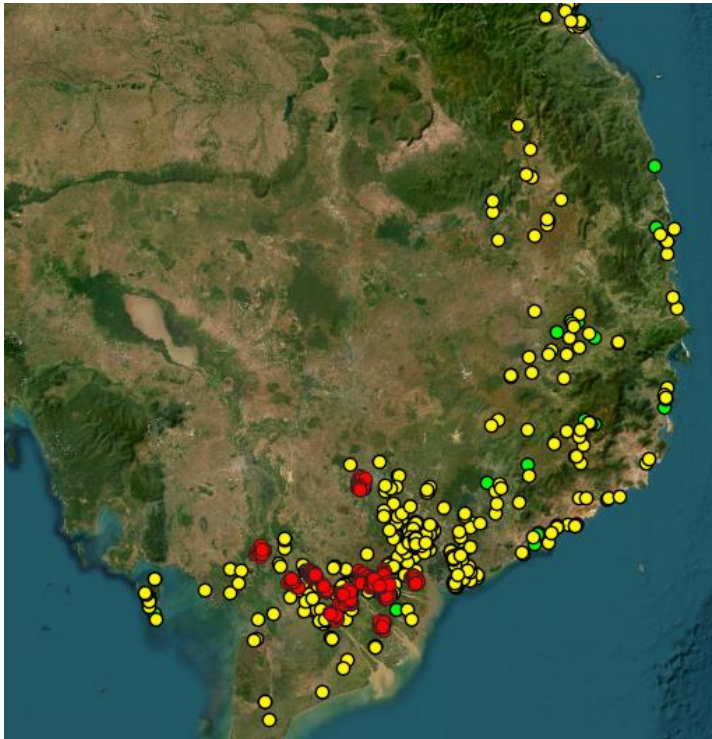
Inverse non-school sampling

- Convert distances to **probabilities of sampling**

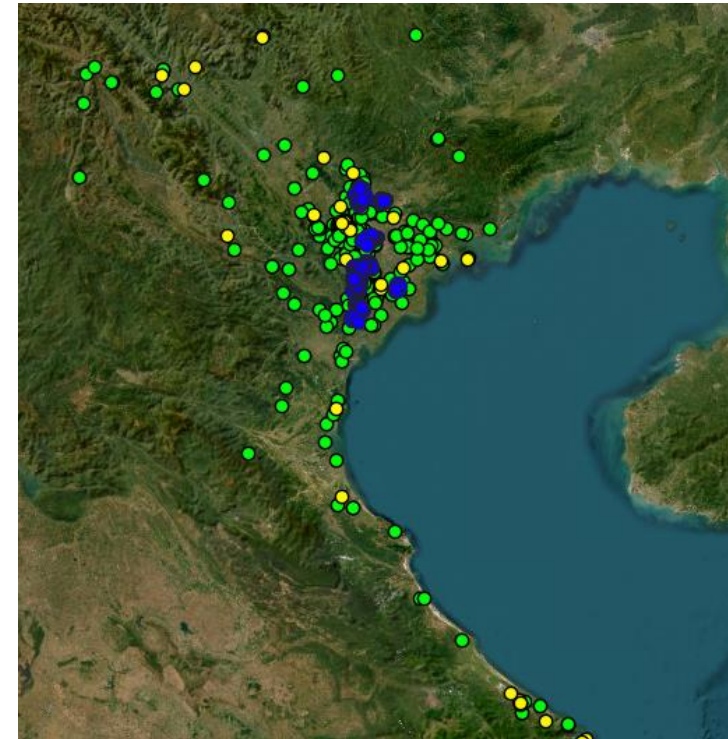
$$p_c(x) = \frac{1}{d_{c,x} + \varepsilon} \quad , \quad \varepsilon = 1 * 10^{-10}$$

- Sample non-schools for each cluster individually (**no repetition**)

Inverse non-school sampling



- - South School Cluster
- - South Sampled Non-Schools



- - North School Cluster
- - North Sampled Non-Schools

Results comparison

- Random non-school sampling
 - **F1: 81.55%**
 - P: 95.94%
 - R: 73.97%
- Inverse non-school sampling
 - **F1: 89.03%**
 - P: 97.38%
 - R: 82.53%

New contributions

Gaussian non-school sampling

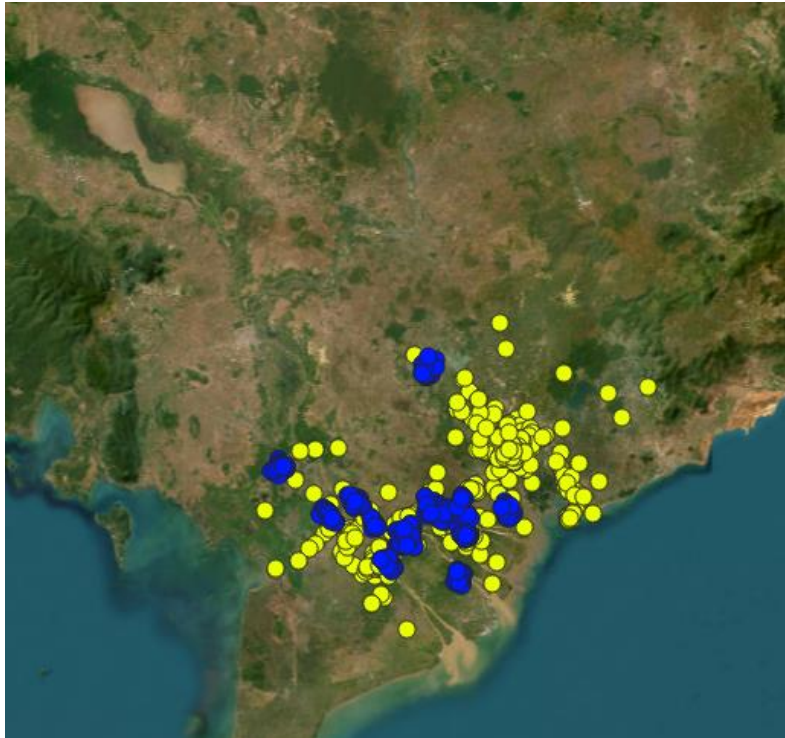
- Schools are split into two clusters
- For each cluster, calculate the corresponding **centroid and variance**
- For each non-school (OSM Vietnam data), calculate the **probability of belonging to a bivariate Gaussian distribution** defined by its corresponding centroid and covariance matrix

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{n/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

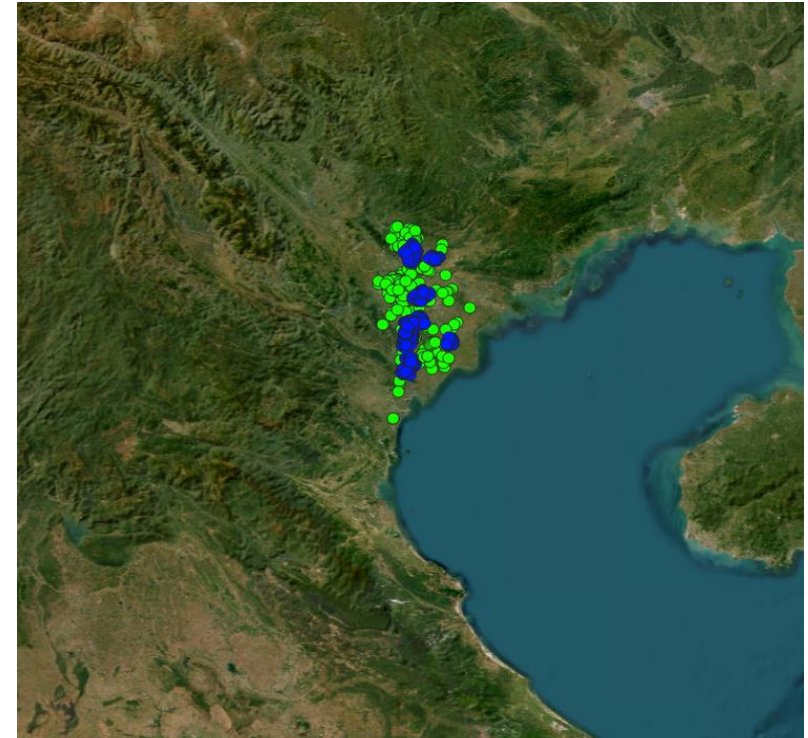
Gaussian non-school sampling

- Use probabilities (normalized bivariate Gaussian PDF) as **probabilities of sampling**
- Sample non-schools for each cluster individually (**no repetition**)

Gaussian non-school sampling

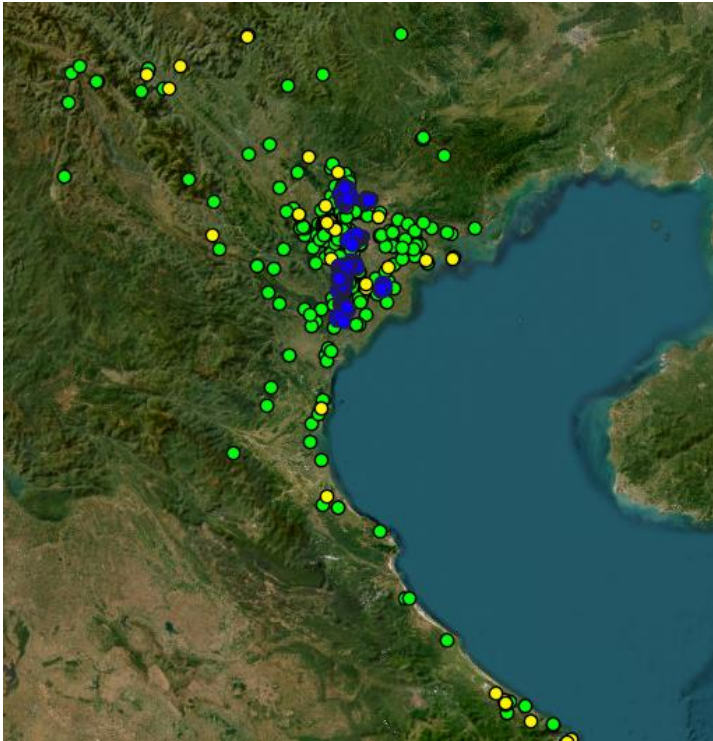


- - School Cluster
- - South Sampled Non-Schools



- - School Cluster
- - North Sampled Non-Schools

Sampling comparison

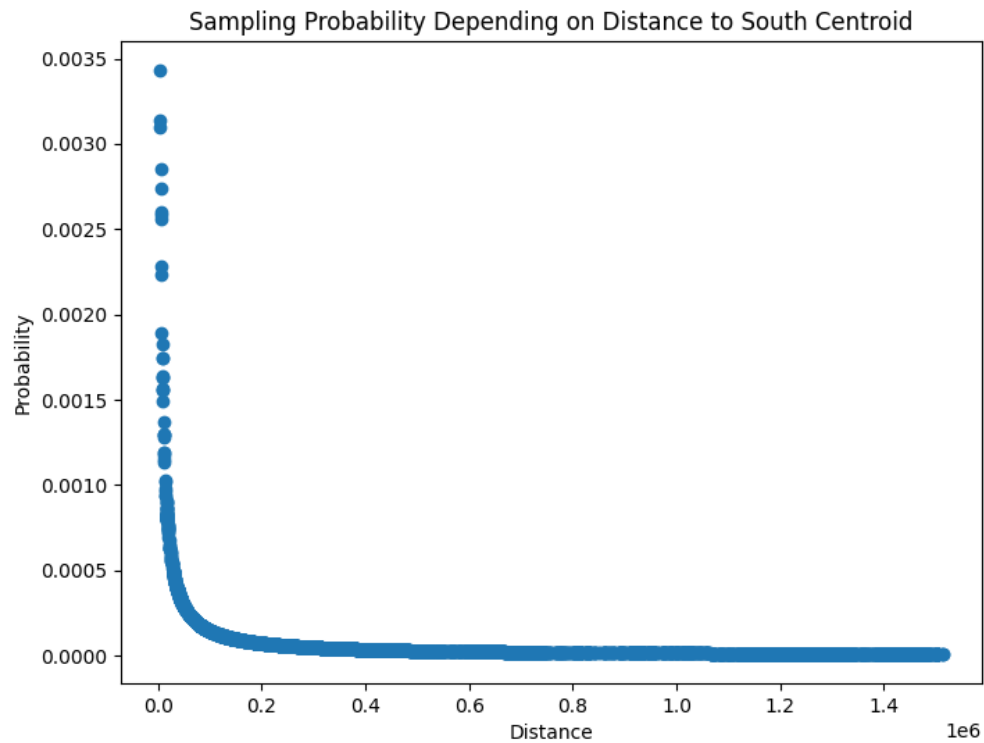


- - North School Cluster
- - North Inverse Sampled Non-Schools

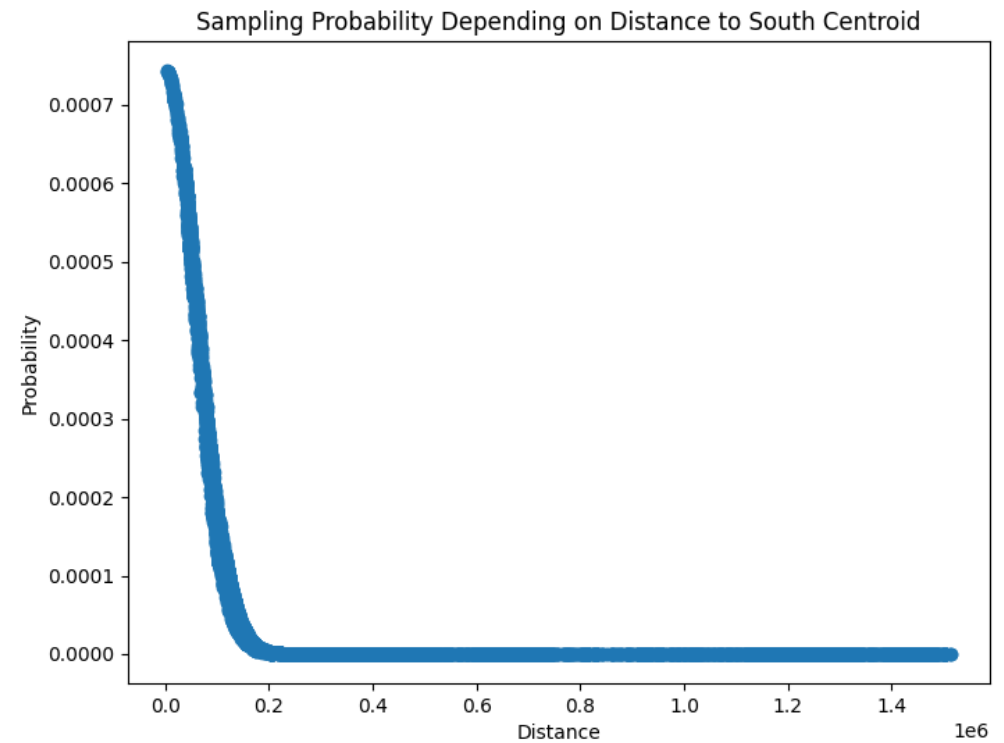


- - North School Cluster
- - North Gaussian Sampled Non-Schools

Sampling comparison



Inverse sampling



Gaussian sampling

Results comparison

- Inverse non-school sampling
 - **F1: 89.03%**
 - P: 97.38%
 - R: 82.53%
- Gaussian non-school sampling
 - **F1: 89.72%**
 - P: 96.04%
 - R: 84.98%

Dynamic non-school sampling

- Previously:
 - Non-schools sampled only once (before training model)
- **Dynamic non-school sampling:**
 - Non-schools sampled independently each epoch
 - Can be used with both inverse and Gaussian sampling

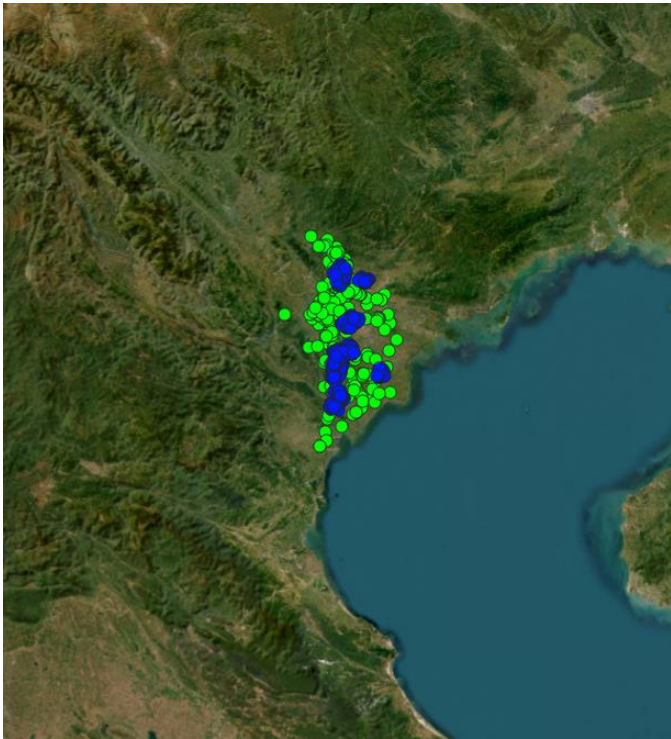
Results comparison

- Static Gaussian non-school sampling
 - **F1: 89.72%**
 - P: 96.04%
 - R: 84.98%
- Dynamic Gaussian non-school sampling
 - **F1: 89.72%**
 - P: 96.32%
 - R: 84.20%

OHEM non-school sampling

- **Online Hard Example Mining:**
 - Each epoch, we evaluate the model on all non-schools
 - For the next epoch, non-schools will be selected:
 - 50% - random examples (inverse or Gaussian)
 - 50% - hard examples
- **Hard example sampling:**
 - Select false positives (non-schools labeled as schools)
 - Use model's confidence score (probability of example being a school) combined with standard (inverse or Gaussian) sampling

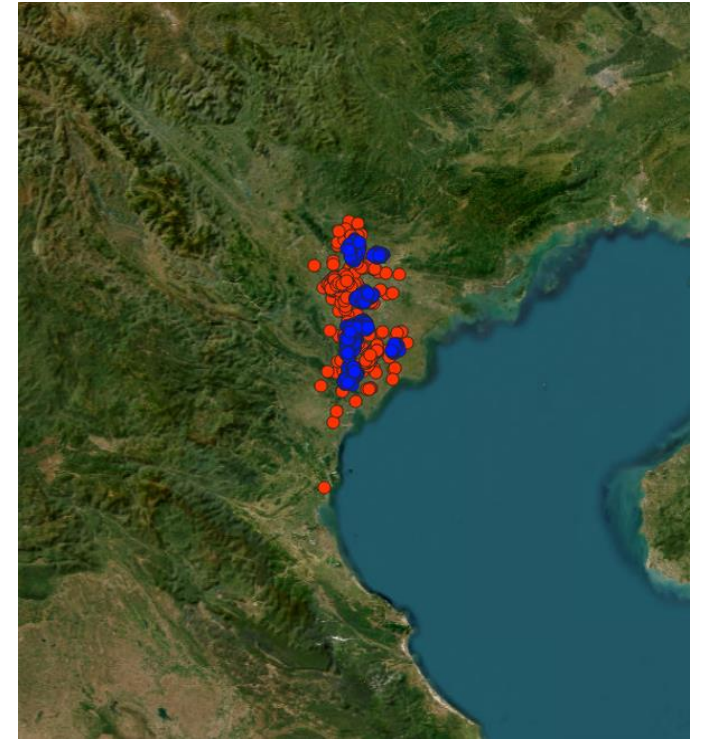
Sampling over epochs comparison



- - North School Cluster
- - Epoch 1 Non-Schools



- - North School Cluster
- - Epoch 5 Non-Schools



- - North School Cluster
- - Epoch 10 Non-Schools

Results comparison

- Dynamic Gaussian non-school sampling
 - **F1: 89.72%**
 - P: 96.32%
 - R: 84.20%
- OHEM Gaussian non-school sampling
 - **F1: 88.93%**
 - P: 97.72%
 - R: 82.15%

Results comparison

- OHEM train takes 3x longer
- Possible explanation for worse results:
 - **Incorrectly labeled examples**
 - A robust model will classify these examples “incorrectly” with a high confidence score, leading to a high probability of being sampled as a hard example

Sampling method choice

- Inverse or Gaussian sampling
- Static or dynamic sampling
- Dynamic or OHEM sampling

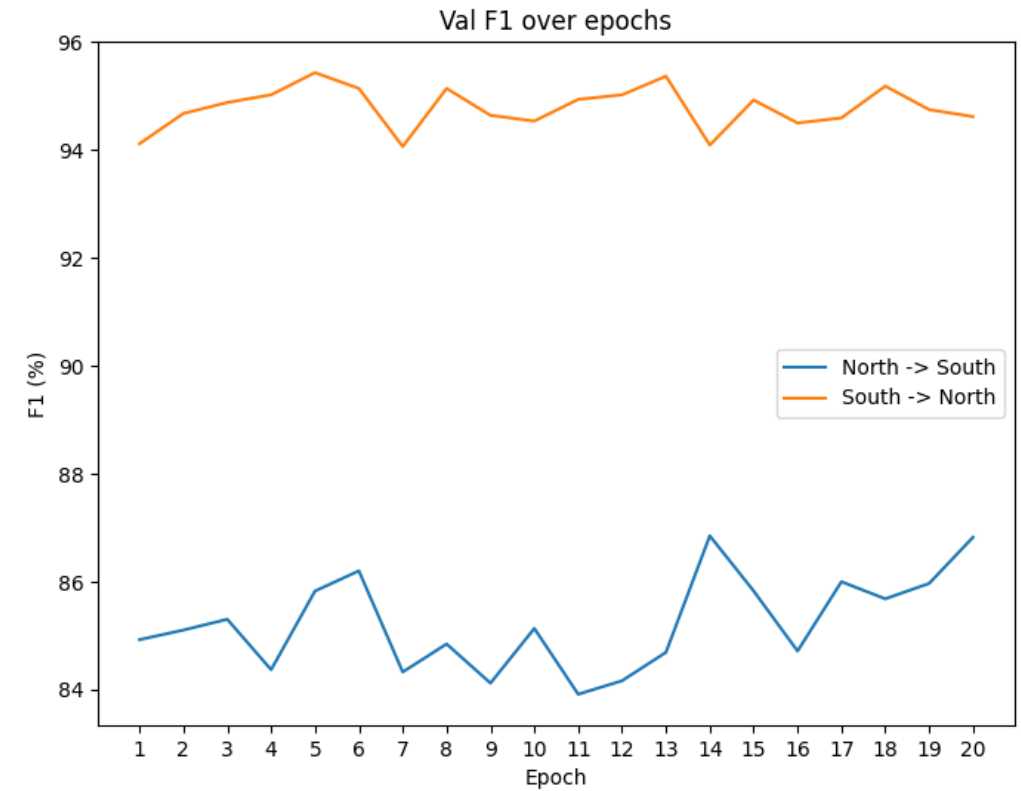
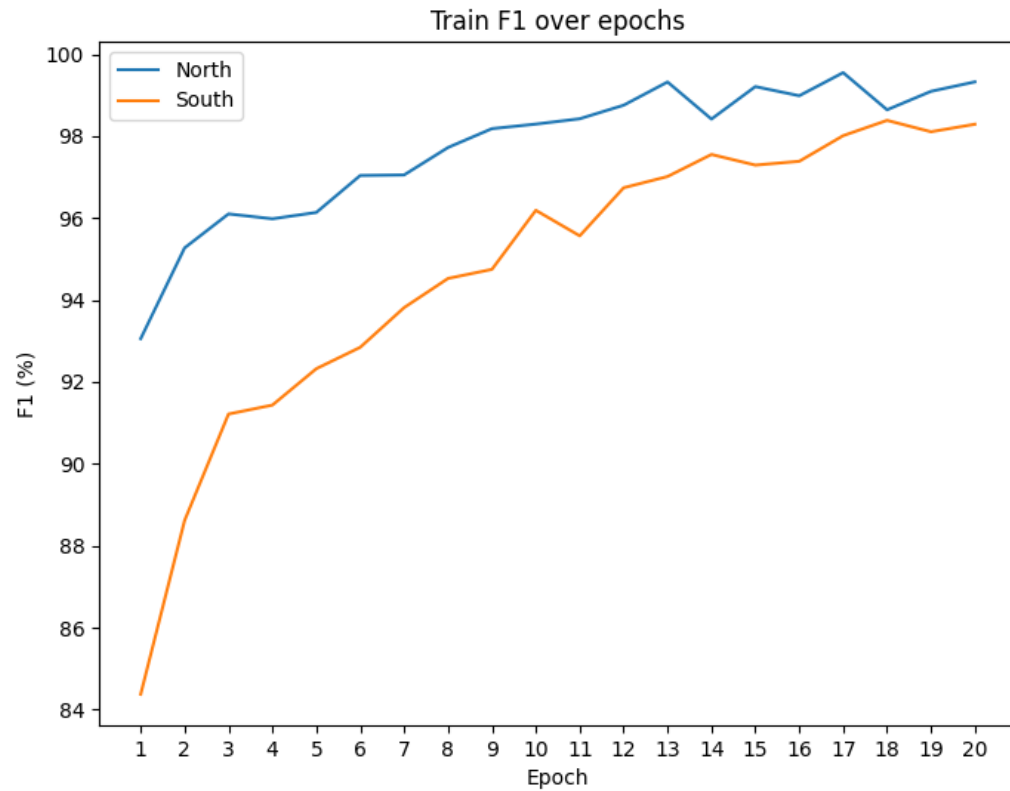
- We've seen that static Gaussian sampling and dynamic Gaussian sampling yield the best results

- Our choice: **static Gaussian sampling**

Number of epochs validation

- Increased number of epochs (20 vs 13)
- Measure **F1-score** over epochs on both train and validation set
- Choose an appropriate number of epochs for fine-tuning

Number of epochs validation



Model fine-tuning

- All Anditi schools + equal number of OSM Vietnam non-schools
- Static Gaussian non-school sampling
- 20 epochs
- Train results
 - **F1: 99.55%**
 - P: 99.80%
 - R: 99.30%

Worst non-schools, train



Worst schools, train



Future work

- Dense inference (run newly fine-tuned model on tiles covering the chosen 26 districts)
- Suggested deadline: **5.12.2024.**